

SHANTI Bill 2025

Transforming India's Nuclear Energy Framework



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PREFACE

India's nuclear journey over the years has been shaped by caution, experience, and the heavy lessons of history. From early ambitions of self-reliance to complex global partnerships, nuclear policy in India has never been merely about energy. It has also been about responsibility, sovereignty, and public trust.

The recently proposed **SHANTI BILL**, introduced as a measure to promote peaceful nuclear cooperation between India and the contemporary world, has reopened some of the long-standing debates. While the government projects it as a progressive step toward clean energy and international trade interwoven with self-reliance, critics warn that it may dilute hard-earned safety norms in favour of foreign commercial interests and may lead to geopolitical unrest.

This research adopts an analytical approach to examine the SHANTI Bill in light of India's nuclear history, global precedents, and present-day concerns. It also attempts to study the SHANTI Bill from multiple perspectives, including the economic, political, and geopolitical arenas. It does not aim to oppose or support the bill blindly, but rather to analyse whether the promise of nuclear "peace" truly balances public safety with international trade interests. Through this study, we seek to develop critical thinking and a deeper understanding of how global agreements affect a nation internally.

THE INTRODUCTION

India's rising energy needs have steered the government to look beyond the domestic market and build foreign partnerships. In this context, the **SHANTI Bill** is presented as a framework to promote peaceful nuclear collaboration, facilitate technology transfer, and smooth international trade relations. On the surface, the bill highlights clean energy expansion, diplomatic trust, and economic growth as its key objectives.

From a **political perspective**, the bill reflects India's broader shift toward strategic alignment with Western powers. By easing regulatory and liability concerns, the government aims to project India as an investment-friendly destination, strengthening executive control over long-term energy policy. Critics, however, argue that parliamentary scrutiny and public debate appear limited, raising concerns about transparency and democratic accountability.

The **geopolitical dimension** of the SHANTI Bill is significant in the context of growing global power competition, particularly in the Indo-Pacific region. Nuclear cooperation with the United States is viewed as a strategic partnership that may strengthen India's international position. However, concerns remain that increased dependence on foreign partners could affect India's strategic autonomy and energy security.

From an **economic perspective**, the bill aims to attract foreign investment and advanced nuclear technology to support India's long-term energy requirements, which domestically is not feasible. While nuclear projects may generate employment, infrastructure growth, and lower

carbon emissions, critics warn that relaxed liability provisions could transfer financial risks from foreign suppliers to Indian public institutions and taxpayers.

In terms of **safety and environmental impact**, questions continue regarding accident liability, nuclear waste management, and supplier accountability. Past industrial disasters and public opposition to nuclear projects highlight the need to prioritize public safety and environmental protection in nuclear policymaking.

This project assesses whether the SHANTI Bill upholds the principles of peace ironically, with the motive after which it has been aimed, and safety, political, geopolitical, and economic interests.

History of India's Nuclear Power and Uproariousness

It was as early as 1901 when the abundant presence of mineral ores in India was discovered. This was done through the Geological Survey of India. Nuclear power stands as the 5th largest source of electricity in India, only after coal, hydro, solar, and wind. As of April 2025, India operates with 25 nuclear reactors in 7 nuclear power plants with an aggregate capacity of 8,880 MW. India is home to rich deposits of radioactive ores, which include thorianite, uranium, uraninite (pitchblende), and other thorium-grounded energies.

In 1946, the Board of Scientific and Industrial Research (BSIR) set up an Atomic Research Committee under Homi J. Bhabha (the Father of the Indian Nuclear Program) to explore India's nuclear energy coffers and to develop and harness them. In 1947, a Uranium Unit was established to concentrate on relating and developing coffers of uranium-bearing minerals. A premonitory board for exploration in nuclear energy was set up. The notable scientists who were necessary in setting up the uranium unit were Daulat Singh Kothari, Meghnad Saha, Homi J. Bhabha, Sir K. S. Krishnan, Darashaw Noshervan Wadia, and Nazir Ahmed.

On 23 March 1948, Prime Minister Jawaharlal Nehru introduced the Atomic Energy Bill in the Indian Parliament, and it was later passed as the Indian Atomic Energy Act. Modelled on the British Atomic Energy Act 1946, the Act granted broad powers to the central government over nuclear wisdom and exploration, including surveying for nuclear minerals, the development of similar mineral coffers on an artificial scale, conducting exploration regarding the scientific and specialized problems connected with developing nuclear energy for peaceful purposes, the training and education of the necessary labor force, and the fostering of abecedarian exploration in the nuclear lores in Indian laboratories, institutes, and universities. With effect from 1 June 1948, the premonitory board for Research in Atomic Energy, together with its parent organization, the CSIR, was folded into the new Department of Scientific Research and placed directly under the Prime Minister. On 3 August 1948, The Atomic Energy Commission of India (AEC) was established and made separate from the Department of Scientific Research, with Bhabha as its first president.

In February 1960, it was decided that the first power factory would be erected in Western India, with locals in Rajasthan, near Delhi, and near Madras noted for unborn marketable reactors. The agreement for India's first nuclear power factory at Rajasthan, RAPP- 1, was

inked in 1963, followed by RAPP- 2 in 1966. These reactors contained rigid safeguards to ensure they would not be used for a military program. RAPP-1 began operation in 1972.

Energy Reserves Around the World

India's domestic uranium reserves are small, and the country is dependent on uranium imports to fuel its nuclear power industry. Since the early 1990s, Russia has been a major supplier of nuclear energy to India. Due to abating domestic uranium reserves, electricity generation from nuclear power in India declined by 12.83 from 2006 to 2008.

An Early Changing

In March 2011, large deposits of uranium were discovered in the Tummalapalle belt in Andhra Pradesh and in the Bhima reservoir in Karnataka by the Atomic Minerals Directorate for Exploration and Research (AMD) of India. The Tummalapalle belt uranium reserves promise to be one of the world's top 20 uranium reserves. 44,000 tonnes of natural uranium have been discovered in the belt so far, which is estimated to have three times that quantity.

Nuclear Agreements with the R.O.W

As of 2016, India has inked civil nuclear agreements with 14 countries: Argentina, Australia, Canada, the Czech Republic, France, Japan, Kazakhstan, Mongolia, Namibia, Russia, South Korea, the United Kingdom, the United States, and Vietnam.

India's nuclear power trip has been guided by a series of cornerstones of legislation that assured the peaceful use of nuclear energy while securing public interests. Each step reflected the country's growing confidence and maturity in managing nuclear technology responsibly.

The Atomic Energy Act, 1962 replaced the earlier 1948 law and laid down the foundation for India's nuclear program. It empowered the government to regulate nuclear energy for peaceful purposes, imposing strict control over exploration, development, and use of nuclear accoutrements.

Amendments in 1986, 1987, and 2015 to the Atomic Energy Act of 1962 gradually opened the sector beyond the Central Government, allowing government companies and common gamblers to share in nuclear power generation. The emendations reflected India's intent to expand capacity while keeping strategic oversight complete.

The Civil Liability for Nuclear Damage Act, 2010, introduced a no-fault liability governance, including compensation in case of nuclear incidents. This law brought clarity on responsibility and erected public trust by prioritizing safety and responsibility in nuclear operations.

Blazoned in the Union Budget 2025-26, it allocates ₹ 20,000 crore to drive the design, development, and deployment of Small Modular Reactors (SMRs).

Target: At least five indigenously designed SMRs to be functional by 2033, strengthening India's clean energy roadmap. • enterprise by Bhabha Atomic Research Centre (BARC) ✓ 200 MWe Bharat Small Modular Reactor (BSMR-200) ✓ 55 MWe (megawatt electrical) SMR-55

55 ✓ Up to 5 MWth (megawatt thermal) High-temperature gas-cooled reactor for hydrogen generation.

IMPACT OF THE SHANTI BILL, 2025 ON THE INDIAN ECONOMY

– An International Comparison

The SHANTI Bill, 2025, has marked a landmark reform in India's civil nuclear sector. By replacing the Atomic Energy Act, 1962, and the Civil Liability for Nuclear Damage Act, 2010, the bill opens nuclear power generation to private and foreign participation for the first time.

Several international experiences from the USA, South Korea, the UK, and Japan have shown that such liberalisation can significantly boost investment and industrial efficiency while also highlighting the importance of strong safety oversight. The following are a few on the same lines.

In the USA, the Atomic Energy Act of 1954 enabled private participation, leading to rapid capacity expansion and technological innovation, thereby creating over 500,000 jobs, ensuring stable electricity prices, and annual tax revenues of around \$30 million per plant. The nuclear industry today is valued at nearly \$40 billion.

South Korea's PPP model, involving state utility KHNP and private conglomerates, reduced costs through standardization, enabled nuclear power to supply about 30% of electricity, and transformed the country into a global nuclear exporter.

Japan's mixed public-private framework reduced oil imports and strengthened industrial growth, though the Fukushima disaster underscored the economic costs of inadequate risk management, leading to higher fossil fuel imports and energy prices—an important lesson for India.

IS THE BILL AN ECONOMIC BREAKTHROUGH OR A STRATEGIC GAMBLE FOR INDIA?

To achieve 100 GW nuclear capacity by 2047, India requires ₹15–20 lakh crore in investment. The SHANTI Bill ends NPCIL's operational monopoly, thereby enabling private Indian firms and joint ventures to pitch in, while allowing up to 49% FDI. This facilitates access to global capital, advanced reactor designs, and safety technologies. Public investment acts as a catalyst, crowding in private capital and reducing fiscal pressure on the government.

A ₹20,000 crore Nuclear Energy Mission promotes Small Modular Reactors (SMRs), which have lower sunk costs, shorter construction timelines, and flexible deployment. SMRs are particularly suited for decarbonizing hard-to-abate sectors such as steel, cement, fertilizers, refineries, and hydrogen production. Localisation of high-precision manufacturing benefits firms like L&T, MTAR Technologies, and MSMEs, generating technology spillovers into defence, aerospace, and advanced materials. Private participation is expected to reduce delays and cost overruns, improving project efficiency.

Nuclear power provides reliable 24/7 baseload electricity, essential for emerging sectors such as AI data centres, semiconductor fabs, and green hydrogen. Expansion reduces dependence on coal, oil, and gas imports, insulating India from global price volatility. Alignment of India's liability framework with international norms improves access to climate finance, green bonds, and multilateral funding, supporting India's net-zero target by 2070. Lower energy imports and higher FDI inflows improve the current account balance, strengthen foreign exchange reserves, and support rupee stability.

Nuclear power contributes to long-term tax revenues, reduces energy subsidy burdens, and generates high-skilled employment in engineering, safety, radiation management, and R&D. Beyond electricity, non-power applications—such as medical isotopes, radiotherapy, food irradiation, and nuclear-powered desalination—enhance economic benefits. Nuclear parks also stimulate regional development through improved infrastructure and ancillary industries.

Critics highlight that capped operator liability and the removal of supplier liability could transfer risk to the public, potentially burdening taxpayers during a major accident. Lessons from Fukushima stress the importance of disaster-resilient design, insurance pools, sovereign backstops, and robust emergency preparedness. Effective implementation requires independent regulation, transparent oversight, and a clear separation between operator, regulator, and policymaker to maintain public trust.

IMPACT OF SHANTI BILL 2025 ON THE ECONOMY

1. Impact of shanti bill on FI

Company/Group	Global Involvement	How SHANTI Could Affect India's Role
Rosatom (Russia)	Major global reactor builder and financier	Partner with Indian companies on reactor technology, component supply, project execution, and financing models
EDF / Framatome (France)	Nuclear operator and technology exporter	Potential reactor supplier and joint-venture partner with Indian nuclear operators
CGN / CNNC (China)	Large-scale reactor builder and global investor	Possible technology supply through Indian partners, subject to strategic and policy clearances
KEPCO (South Korea)	Exported reactors globally (e.g., UAE)	Engineering, construction, testing, and technology partnerships with Indian firms
Westinghouse / GE Hitachi (US–Japan)	Large reactors and SMRs worldwide	Supplier of reactors, SMR technology, fuel-cycle services via licensed Indian entities
Canadian nuclear firms	Fuel, uranium, and nuclear components	Fuel supply and component partnerships through Indian licensees
Global investment funds	Financing nuclear and infrastructure projects	Capital inflows into Indian nuclear projects through public–private partnership models

2. Fiscal impact

Outline: How electricity generation impacts the Balance of Payments

India cannot export electricity easily, so higher electricity production does not directly earn foreign exchange. When electricity generation relies on imported coal and oil, it leads to higher imports, increasing dollar outflows and widening the trade deficit. A larger trade deficit puts pressure on the rupee and strains government finances, often forcing the government to increase energy subsidies or spend more on price support. In contrast, when coal and oil imports are reduced, foreign exchange outflows decline, the trade deficit improves, and the government saves on import bills and energy subsidies, strengthening both the fiscal balance and the current account position. The bill helps India in not just reducing the

dependence on imported oil and coal but also on saving imports of coal and oil, which by itself improves fiscal health.

3. India's Energy Import Dependence

India's energy sector has historically been import-dependent, particularly for coal and crude oil. The **Shanti Bill** aims to reform the electricity and energy ecosystem by improving efficiency, encouraging cleaner energy sources, and reducing excessive dependence on fossil

fuel imports. Import trends before and after the policy provide useful insights into its impact.

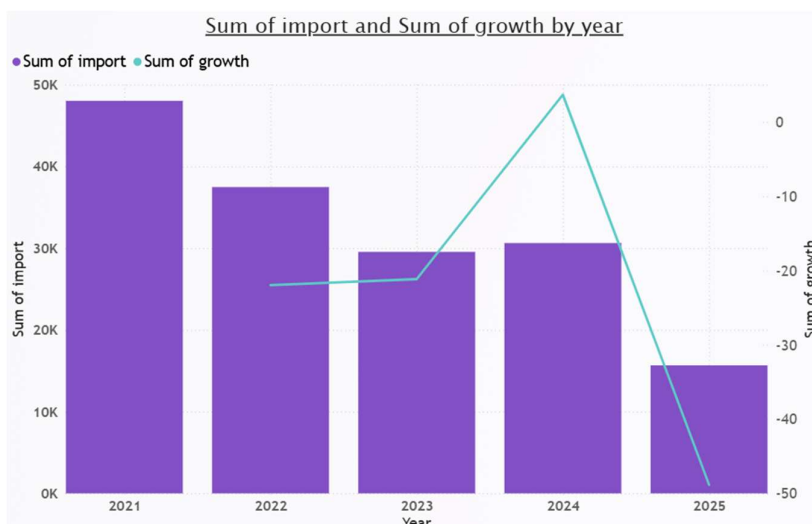
1. Coal

The declining coal import trend after 2021 indicates improved domestic coal availability and a stronger focus on energy efficiency and alternative power sources. The temporary increase in imports during 2024 reflects short-term demand pressures and transitional dependence during

the restructuring of the energy sector. The sharp contraction in 2025, along with negative growth, suggests that structural reforms under the Shanti Bill are beginning to reduce coal import dependence.

Policy linkage

The Shanti Bill promotes better power-sector management, reduced transmission and distribution losses, and diversification of energy sources. Together, these factors lower the need for imported coal, particularly for electricity generation.



2. Oil

Rising oil imports until 2022–23 reflect global oil price shocks and post-pandemic demand recovery. The slowdown and decline after 2023 indicate reduced growth in oil consumption, improved energy efficiency, and partial substitution through renewables.

Policy linkage

Although the Shanti Bill does not directly regulate oil imports, it encourages electrification and

reduces fossil fuel dependence in power generation. This indirectly curbs long-term oil import growth, especially in the energy and transport sectors.

Economic Implications

Reduced imports lead to a greater emphasis on domestic energy production and job creation in renewable energy, grid management, and energy services. Import substitution under the Shanti Bill supports domestic industries and generates skilled and semi-skilled employment.

Short-term effects

- Temporary import fluctuations due to demand shifts and global price volatility.
- Adjustment costs during the transition phase.

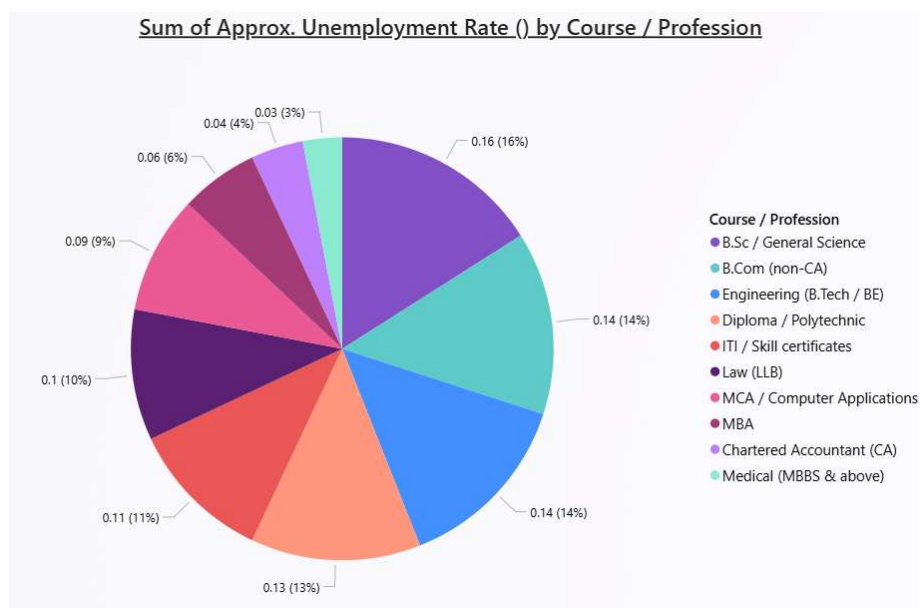
Long-term effects

- Sustained decline in fossil fuel imports.
- More sustainable and predictable energy sourcing and Stronger energy security with improved external stability.

4. Jobs and Employment

Educated Unemployment Pattern (PLFS)

The distribution shown in the Periodic Labour Force Survey indicates that graduates and diploma holders account for the largest share of educated unemployment, despite India's low aggregate unemployment rate of around 3.2%.



Impact of the Shanti Bill on Skilled Labour

The Shanti Bill promotes electricity sector reforms, efficiency improvements, grid modernisation, and renewable energy integration. These reforms are expected to increase demand for skilled professionals, particularly electrical and mechanical engineers, automation and

control engineers, and energy management specialists. Since engineering graduates constitute nearly 14% of the educated unemployed, the bill has the potential to absorb a significant portion of this workforce in the medium term.

→ **Role of Diploma and ITI Holders**

Diploma holders (13%) and ITI-trained workers (11%) together form a substantial share of the unemployed educated workforce. The Shanti Bill may raise demand for technicians, machine controllers, and maintenance personnel while encouraging skill upgrading and reskilling. However, rising automation may reduce demand for low-skilled manual roles, making continuous skill development essential.

→ **Labourers and Informal Employment**

A large share of labourers remains engaged in informal, casual, or contract-based employment, which keeps official unemployment low. The transition away from traditional energy systems may displace workers in coal-based and manual operations, potentially leading to underemployment rather than open unemployment if adjustment policies are absent.

→ **Structural Shift in Employment**

The Shanti Bill represents a structural transformation of the labour market by shifting workers from low-productivity informal roles to more productive, capital- and skill-intensive energy and infrastructure sectors.

Overall Employment Impact

While the Shanti Bill is unlikely to alter India's headline unemployment rate significantly, it can improve the structure and quality of employment by reducing educated unemployment, particularly among engineers and diploma holders. Its success will depend on effective reskilling, inclusive transition strategies, and strong alignment between education and industry needs.

BALANCING NUCLEAR EXPANSION WITH SAFETY AND SOVEREIGNTY:

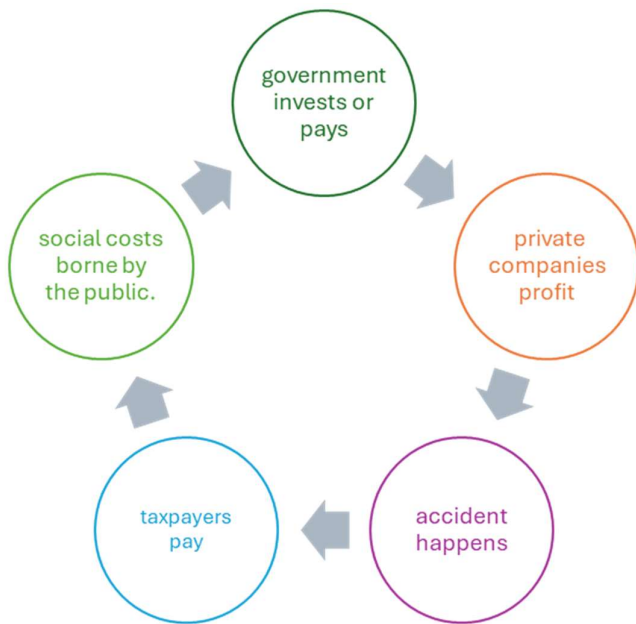
While the shanti bill marks a potential inflexion point in the history of the nation's atomic and nuclear sector by opening it up to greater private participation, the quantum of privatization and liberalisation involved in it in lieu of attracting private investments needs to be understood. This segment argues that without robust safeguards, the liberalisation envisaged under the SHANTI Bill risks transferring systemic risks from private actors to the state while weakening India's long-standing state-led nuclear strategy.

Consider the model of a big power plant being established by the government which lays the foundation and provides land, infrastructure and regulatory support so the project can attract foreign companies. These companies build and run the plant, making profits from selling electricity and other products.

Now, if something goes wrong say a reactor malfunction happens or equipment failure takes place the private companies' liability is capped, and suppliers aren't responsible at all. That

means the government has to cover the expenditure for the damages, cleanup, and compensation all of which are ultimately funded by the Taxpayers.

So, this may turn out into a loop



This may turn never ending if the government CONTINUOUSLY INVESTS in infrastructure and subsidies to attract private nuclear participation, while private companies profit under limited liability. If an accident occurs, taxpayers bear the costs, yet the plant continues operating, creating a cycle where public funds repeatedly support private gains.

And with that lets move on to study a few more implications of the bill on the safety and the sovereignty of the country

Does the polluter pay henceforth?

No. The SHANTI Bill significantly alters India's nuclear safety framework by redefining liability and oversight mechanisms in order to facilitate private sector participation. By

removing supplier liability and introducing capped operator liability, the Bill weakens the "polluter pays" principle that underpinned India's earlier nuclear liability regime. In the event of a serious nuclear accident, this structure of the new bill risks transferring long-term health, environmental, and

categories of nuclear installation

Reactor Thermal Capacity	Liability Cap (₹ Crore)
Above 3,600 MW	₹3,000 crore
1,500 MW – 3,600 MW	₹1,500 crore
750 MW – 1,500 MW	₹750 crore
150 MW – 750 MW	₹300 crore
Below 150 MW (and fuel processing units)	₹100 crore

rehabilitation costs to the state and affected communities rather than holding profit-making entities fully accountable removing the "right to recourse" against suppliers. Critics argue that it might socialize the risk since the union government assumes residual liability if the liability cap exceeds the setting thereby transferring, financial pressure on the state and ultimately the tax payers

Transparency:

Nuclear energy requires exceptional transparency due to its potential for irreversible and catastrophic harm. Provisions in the SHANTI Bill allowing the government to classify wide categories of nuclear information as restricted limit public access to safety audits, operational details, and accident-related information. This limits and constraints the role of media, and independent experts in monitoring safety compliance and measures. Reduced transparency may weaken public trust on the government and stagger the early detection of safety laps, which is very essential in a high-risk sector which is now privatized and open to the ROW.

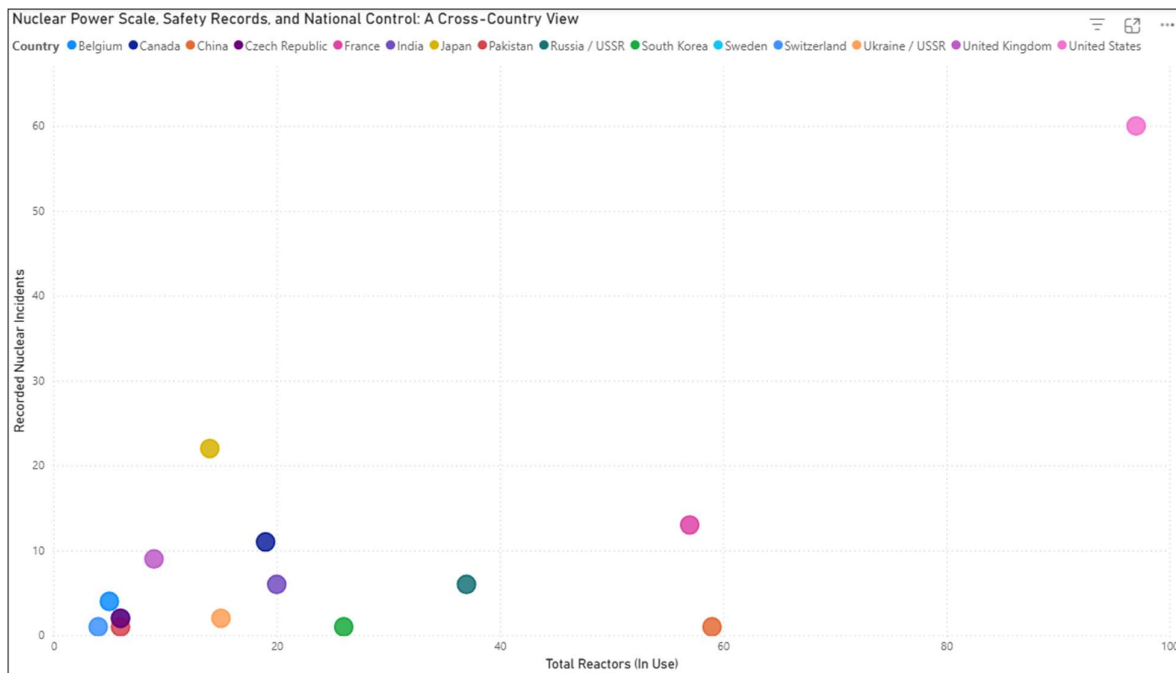
Sovereignty and Strategic Control:

The opening of nuclear power generation to private and foreign-linked entities raises important sovereignty concerns in a sector traditionally viewed as a core state responsibility.

Country	Recorded Nuclear Incidents	Sovereignty Code	Total Reactors (In Use)
Japan	22	1	14
United States	60	1	94
Belgium	4	2	5
Canada	11	2	19
Sweden	2	2	6
Switzerland	1	2	4
United Kingdom	9	2	9
Czech Republic	2	3	6
France	13	3	57
South Korea	1	3	26
Ukraine / USSR	2	3	15
China	1	4	59
India	6	4	20
Pakistan	1	4	6
Russia / USSR	6	4	37

(table showing countries and their sovereignty codes along with the accidents per reactors)

Critics argue that profit-driven participation in such a sensitive domain could erode state control over technology choices, project execution, and long-term strategic planning. There are fears that foreign vendors may influence reactor design and fuel decisions, potentially sidelining India's indigenous three-stage nuclear programme and long-term thorium ambitions, which are central to national self-reliance.

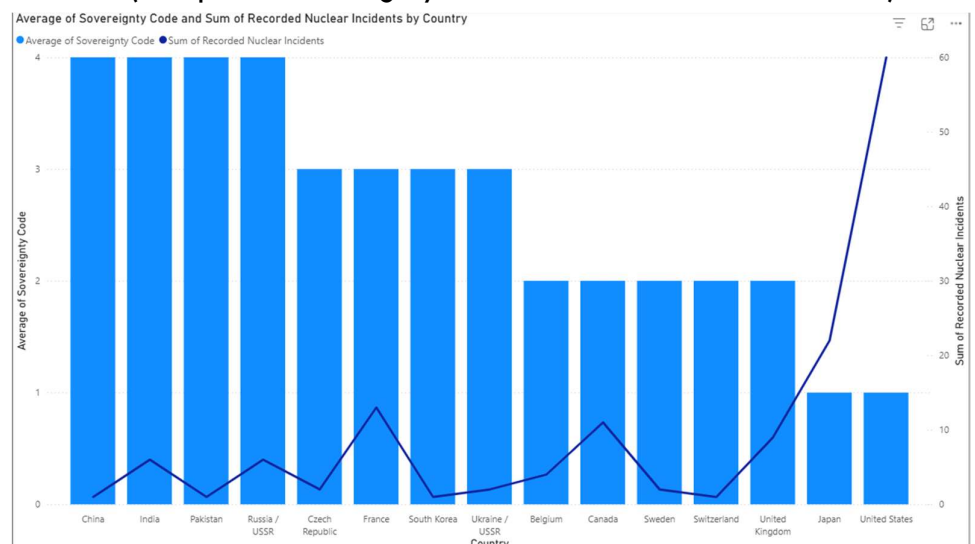


(Graph 1: nuclear power scale and ACCIDENTS)

As shown in the first graph plotting total reactors against recorded nuclear incidents, countries that have expanded nuclear capacity rapidly such as the United States, France, China, and Russia occupy the upper-right region, indicating that higher reactor numbers are accompanied by higher absolute incident counts. This pattern reflects scale-induced exposure rather than weak safety systems. India's location on the graph, with a moderate number of reactors and a low incident count, visually demonstrates a cautious expansion strategy under tight state control. The contrast suggests that India's sovereign, state-led model limits operational exposure by prioritising strategic oversight over rapid capacity growth, reinforcing the argument that nuclear energy expansion is inseparable from questions of state capacity and control.

(Graph 2: sovereignty code and accidents of countries)

The second graph, which juxtaposes average sovereignty control with total recorded incidents, further sharpens this relationship. Countries with high sovereignty scores such as India, China, and Russia cluster on the graph with lower cumulative incidents, while countries with lower



sovereignty scores, notably the United States, Japan, and the United Kingdom, display steep rises in incident totals. If India were to liberalise its nuclear sector and move toward a lower

sovereignty profile similar to these countries, the graphs suggest a likely shift toward faster expansion accompanied by higher systemic exposure. Over time, private and foreign participation could influence reactor design, fuel sourcing, and operational priorities, potentially sidelining indigenous long-term programmes like the three-stage nuclear cycle. Thus, the visual evidence indicates that reducing sovereign control may transform nuclear power in India from a strategically governed sector into a commercially driven one, with long-term implications for autonomy and national security.

Technological and Data Sovereignty:

Nuclear power plants rely heavily on advanced digital systems for control, monitoring, and safety management. Increased reliance on imported technologies and software may create dependencies that limit India's control over upgrades, maintenance, and emergency response which might go ironically against strengthening civil dependence on the domestic sector just with a little help of FDI and their participation. Additionally, nuclear energy data including operational performance and consumption patterns has strategic significance. If not governed by strong data localisation and security norms, such information could be vulnerable to external access.

While expanding the nuclear sector for investment is essential for India's long-term energy security and decarbonisation goals, the manner of expansion is equally critical. liberalisation should be interwoven with scrutiny and strong liability at all levels to ensure that it does not just reduce dependency but also make sure that the sovereignty of the country is maintained. hence it is necessary to ensure that nuclear expansion of the country strengthens rather than compromising national interest.

INDIA'S NUCLEAR PRIVATIZATION AND THE GEOPOLITICS OF NON-ALIGNMENT

INTRODUCTION:

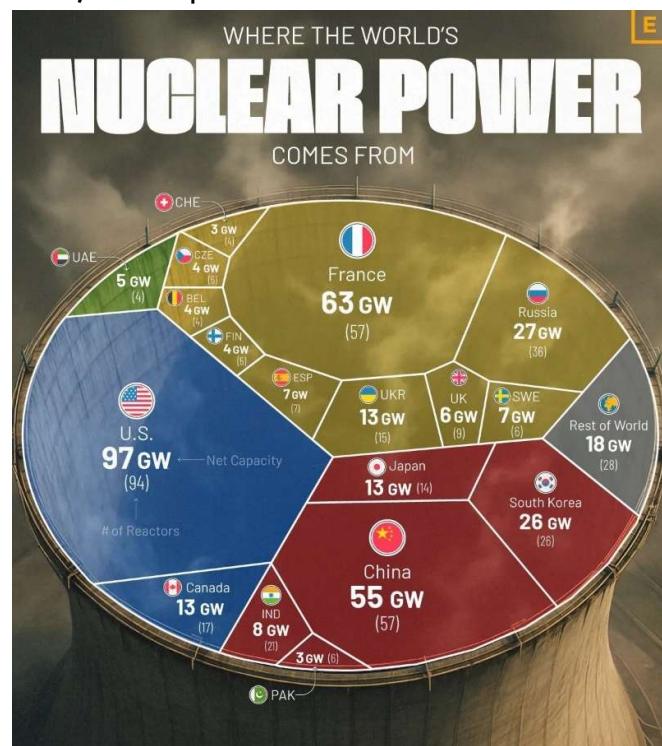
The current international system is defined by strategic uncertainty, where economic interdependence coexists with deep geopolitical mistrust. Energy, once treated as a purely developmental concern, has re-emerged as a decisive instrument of power shaping alliances, rivalries, and long-term influence. In this environment, nuclear energy occupies a unique position at the intersection of national security, technological sovereignty, and global diplomacy. For countries seeking rapid growth without compromising climate commitments, nuclear power has become not just an option but a strategic necessity. India's evolving nuclear policy must therefore be viewed not in isolation, but as part of a wider contest where technology transfer, investment flows, and regulatory frameworks increasingly reflect geopolitical alignments.

As India advances ambitious reforms in its nuclear energy sector, the implications extend far beyond electricity generation or infrastructure development. Deeper engagement with foreign technology providers and private capital raises critical questions about strategic autonomy, regulatory control, and long-term dependency in a multipolar world. Where does cooperation end and influence begin? Can a developing power integrate into global nuclear supply chains without replicating older patterns of economic and technological

dependence? At a time when several resource-rich states have faced destabilization amid external pressures, the central question confronting India is whether its nuclear transformation will reinforce sovereign decision-making or expose new vulnerabilities under a modern form of geopolitical and economic leverage.

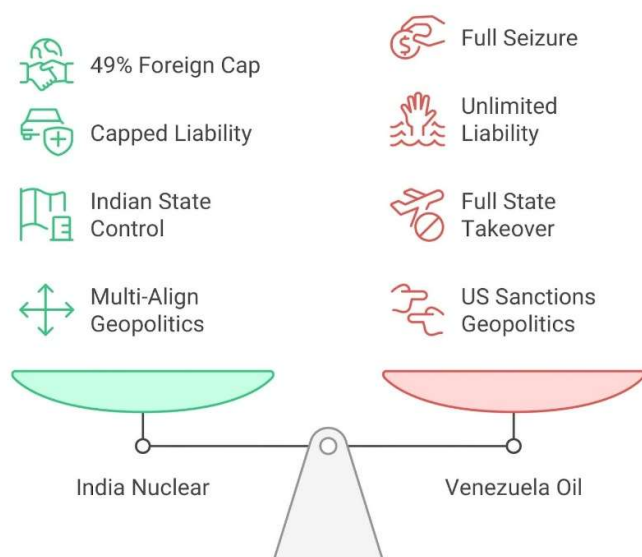
In a world where powerful states exert influence through economic pressure, military action, and control over strategic resources, to what extent might India's deep engagement with foreign powers in critical sectors like nuclear energy risk replicating patterns of neo-colonial dependency, similar to how recent events in Venezuela — including external intervention and competition over oil reserves — have reshaped its sovereignty and policy space?

Venezuela's oil crisis followed a certain course and that course was not forecasted by the higher authorities. Nobody predicted this to happen in the future as they took the decision based on their own self-interest and overlooked the Geopolitics sphere. Firstly, they had



private firms for oil drilling from their land, like how India is opening the nuclear energy sector for private players for the sake of more investment and the technical support that a

Comparing Investment Risks in India and Venezuela



country needs when they found things difficult to setup (technical things). Now coming back to Venezuela after they set up things fully the oil extraction process then they nationalized the oil exploration firms in the mid-1980s seized all of the assets. Which then caused serious problems such as company asking for compensation. They didn't allow the American firms to do the further extraction and take their own amount of crude oil and at own price. This recently U.S president Donald Trump claimed "They stole our oil!". They stole America's oil, which Trump painted as expropriation rather than a sovereign policy of Venezuela. The US claimed the narrative of the US first.

Now the point why we are discussing here as this the hot topic all around the globe the same will not happen with India as they kept a strict cap on investment for foreign firms and now first the nuclear energies generated will be consumed by the own country and it won't be exported. India claims that they are adjusting the regulatory terms within our sovereign jurisdiction as the energy supply is state controlled. If Venezuela shows how resource politics can unravel under external pressure and unpredictable leadership, the Trump era reminded the world that global power today often moves on impulse as much as strategy. Sanctions, tariffs, and policy shifts announced overnight turned uncertainty itself into a geopolitical weapon. In such a climate, strategic sectors rarely remain purely economic for long. India's SHANTI Bill therefore enters a world already primed for competition, where major power's view investment not just as cooperation but as leverage. Understanding this moment requires moving beyond outcomes to examine intentions. The following section analyzes how the United States, Russia, and Japan each approach India's nuclear opening through their own strategic and investment lenses.

MULTI-POWER COMPETITION & INVESTMENT STRATEGIES

The global nuclear market is now fragmenting along geopolitical aspects, with each power positioning India differently:

United States: Commercial Deepening + Conditionality

The US has endorsed its 2008 123 Civil Nuclear Agreement through SHANTI, signalling openness.

US Investments and Commitments:

- US has removed three Indian research entities (BARC, IGCAR, IRE) from the Entry List in January 2025, enabling easier transfer of technology.
- Westinghouse contracted to build 1,208 MW AP1000 units at Kovvada (Andhra Pradesh) site.
- Holtec International approved for Small Modular Reactor(SMR) technology transfer to India under 10- year license. Additionally, it has confirmed to established heavy manufacturing facility in Maharashtra which will reduce the dependence on imports and lead to production of reactor locally in India.
- Trump-Modi February 2025 meeting reinforced the COMPACT for the 21st century initiative, including nuclear energy as one of its pillars.

Russia: Localization as Strategic Depth

Russia positions its nuclear partnership differently: as industrial partnership, not only equipment supply.

Kudankulam & Expansion Plans:

- Units 1 and 2 operational since 2013-2016; Units 3 and 4 expected December 2026; Units 5 and 6 by December 2026 and September 2027.
- Experienced 73% cost overruns and 72-month delays on Units 3-4, with component supply disruptions due to the tension between Russia and Ukraine.
- Rosatom plans to build VVER-1200 reactors (Newer Russian Powerful reactor) for a second site in India.
- Rosatom committed to manufacturing reactor components, fuel assemblies, and spare parts *within India* for future units. This action will start supply chains in India which will remain untouched by sanctions and Geopolitical tensions.

STRATEGIC LOGIC:

By starting to make nuclear parts and fuel inside India, Russia is no longer just foreign company selling reactors and other required equipment. It becomes a long-term industrial partner strongly involved in India's nuclear system. If India ever tried to suddenly replace Russia or take over these projects, Russia could respond by stopping the supply of required components essential for six Kudankulam reactors that already exist. This gives Russia an advantage from within, not from outside. Also, this strategy which Russia is following to make nuclear equipment in India, perfectly fits with India's MAKE IN INDIA goals. Because this

creates a job opportunities and support local manufacturing. **Japan: Technology Transfer via GE Hitachi**

Japan's approach is cautious but technology driven.

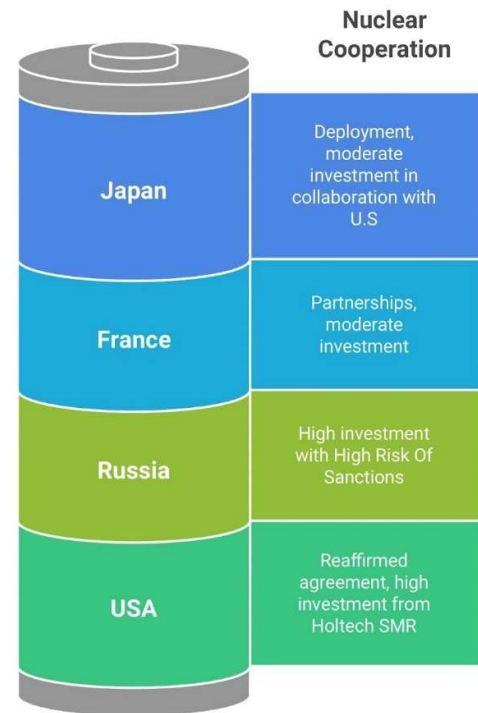
Current Positioning:

Japan is offering india advanced but carefully tested nuclear technology through GE Hitachi. Its small reactor design, BWRX-300, primarily focus on safety, flexibility and cost control. After the government approved an agreement on 2017, which remove the restriction from sharing the nuclear technology, Japan is now demonstrating this technology in Canada before offering it to india. This careful and prof first approach of Japan fits well with India's plan to expand nuclear power safely.

Constraint:

Japan does not have the powerful political influence over India that some other countries do like China, which has ability to block India at Nuclear Suppliers Group (NSG) and Russia, which has the strong strategic partnership with india, especially in defence and nuclear sectors. Due to this, Japan cannot influence the Indian government decisions and remains mainly in technical aspect, not political. Therefore, Japan becomes an attractive and low risk partner along with safe and modern technology.

Country reactions to nuclear energy cooperation range from positive to negative.



US SANCTION GUILLOTINE:

Now coming a little more out of the context let us check the reality of actually how far nuclear energy can go as the very old friend of ours is firing the sanctions like some free bus ride service. Under Trump's administration the **Graham-Blumenthal Bill (2026)**. It says that whichever country purchasing Russian oil, uranium and other important critical minerals will be endorsed up to 500% tariffs. In simpler terms the importer has to pay 5 times the import value of the importing goods. This is thing is even more specified that countries which are "knowingly" purchasing this should pay the tariffs. The tariffs are based on the vibes that trump was experiencing at that point of time. The Make America great again species few countries like India, Brazil, and China quoting the rates at they are purchasing Russian oil as cheap.

Now if they enforce the same on the uranium imports, the reactors which is being supplied by India's Rosatom branch this will become really vulnerable.

To secondary sanctions which is beyond tariffs and it also blocks several things including properties and financial access to involved entities. US has already planned in long term that they are not going to seize only the reactors but also to cut off the fuel supply to these regions of nuclear energy generations which are being operated by Russian agencies like in Kudankulam units 3-6.

This part is not merely classical colonialism instead it is economic coercion via supply chain disruption. Actually, here diplomatic pressure comes to India as we are more dependent on US and global markets than US on India.

Now the US is clearing out things. The US wants you to be either this side or that side, no non-aligned alliance or neutral country like fuss. Mr Trump wants either you to support him in your fullest thought, word and deed. So he initiated a new Silicon Peace framework where it assures to create a trusted AI covering the rare earth metals, energy (which are mostly blocked by the communist power i.e. China) The main goal here is that he wants everybody who is with the peace pact to invest in mines, factories, data centres and optical cables. Now why this new thing as India excluded itself from joining this so called "Pax Silica" as India didn't see future in it. It is just that Trump is suspecting BRICS+ convention and wants something greater against it.

If we see this from a broader perspective India needs rare earth metals as freshly, they allocated Tata and other micron plans for indigenous semiconductor. So, most of the rare earth metals lie with China i.e almost 90% and it will take a lot of time for the US to take the immediate investment on paper to construct real factories and sell out tangible chips. So, it is not the word investment that makes everything happen magically and India needs rare earth materials to make chips. If we go in this AI race it will go on and on. This is an economic weapon which is dangerously planned by the US. That is if you exclude then you miss out the trillion-dollar AI opportunity and a growing part of the world with peace as the main focus [He will get the Nobel Prize]. Else you face the heavy consequence of Graham-Blumenthal 500% tariffs and further sanctions.

In a nutshell after all these points the diplomatic condition of India is very complex. India should take some steps by thinking about it from all angles. India is now being forced into a false binary to choose one of the best friends i.e align with US and abandon Russian partnerships and the discounted energy costs OR ignore Trump (PM Modi's best friend) and face the deadly sanctions and isolate the trillion-dollar western tech opportunity.

To come out of this situation SHANTI Bill can be a dynamic yet a multisource attempt to escape this trap. This bill makes India a very attractive destination for investment as this is happening in such a big country with all the support being subsidised and further additional benefits and that too in an energy generating sector like nuclear.

IMPACT ON GLOBAL SHARE MARKETS

The SHANTI Bill 2025 (Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India) is a landmark legislation passed in December 2025.¹ It fundamentally shifts the Indian nuclear sector from a state monopoly to a public-private partnership model.²

In the share market, this has triggered a "re-rating" of companies in the power, engineering, and capital goods sectors

Key Impacts on the Indian Share Market

The bill introduces several structural changes that have improved investor sentiment:³

- **Ending State Monopoly:** For the first time, private Indian companies can build, own, and operate nuclear power plants.⁴
- **FDI & Private Investment:** It allows up to **49% investment** (foreign or domestic) in nuclear projects, opening doors for massive capital inflows.⁵
- **Liability Reform:** By removing "supplier liability" (which previously made equipment makers legally vulnerable for decades), the bill has significantly lowered the risk for manufacturing companies.⁶

Grid Stability: The bill focus on "baseload" power (24/7 electricity) is seen as a major boost for the reliability of the national grid, benefiting the broader industrial economy.

1. The "Hint" in the Language

- **The Exact Words Used in Parliament**
- **The most critical "hint" came during the section on Energy Transition. The Finance Minister stated:**
- **"Development of at least 100 GW of nuclear energy by 2047 is essential for our energy transition efforts. For an active partnership with the private sector towards this goal, amendments to the Atomic Energy Act and the Civil Liability for Nuclear Damage Act will be taken up."**
- **She also announced the "Nuclear Energy Mission" with these words:**
- **"A Nuclear Energy Mission for research & development of Small Modular Reactors (SMR) with an outlay of ₹20,000 crore will be set up. At least five indigenously developed SMRs will be operationalised by 2033."**
- **For Investors:** This was the signal for them that something was changing related to this ACT. The Atomic Energy Act of 1962 was the only thing preventing private players from entering the sector. Mentioning an amendment meant the government had already decided to end the state monopoly.
- **For Companies:** It signalled that they should stop looking at nuclear as a "government-only" business and start preparing their balance sheets for private partnerships.

2. The Pre-Bill Market Reaction (Feb 2025 – Dec 2025)

The initial spark didn't come from the Bill itself, but from the Union Budget 2025, where the government allocated ₹20,000 crore for Small Modular Reactors (SMRs).¹

- **Valuation Rerating:** Stocks like Walchandnagar Industries and KSB Ltd saw their Price-to-Earnings (P/E) multiples expand. Investors shifted from valuing them as "traditional engineering firms" to "high-tech energy plays."

- "Buy on Rumour": Many of these stocks surged 15–20% in the 30 days leading up to the Bill's introduction, as leaks about "private sector entry" and "liability removal" became more frequent.
- The "Nifty CPSE" Push: Shares of BHEL and Power Grid gained momentum early as they were seen as the "safe" government bets that would anchor any new nuclear mission.

3. Immediate Impact on Stocks (The "Feb-March 2025" Rally)

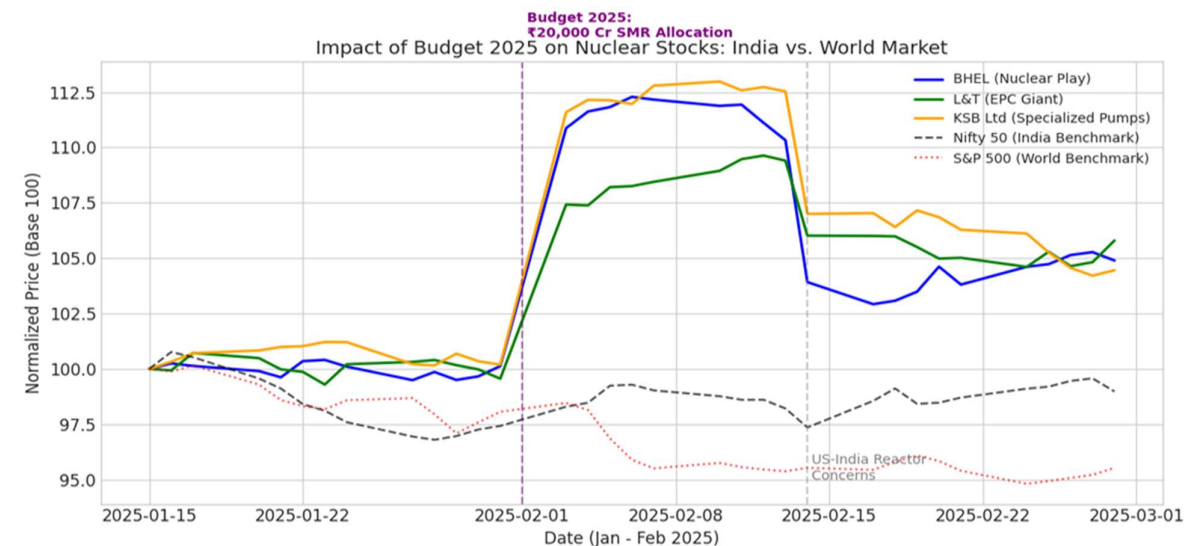
As soon as the ₹20,000 crore **Nuclear Energy Mission** was announced, the "Big 6" nuclear stocks saw an immediate surge, often hitting "upper circuits" (maximum daily gain allowed):³

Stock	Initial Reaction (Post-Budget)	Market Sentiment Hint
BHEL	Rose ~8-10%	Investors realised BHEL would be the lead partner for the new "Bharat Small Reactors" (BSR).
L&T	Gained ~5%	Seen as a "safe harbour" bet for large-scale EPC (Engineering, Procurement, Construction) contracts.
KSB Ltd	Rose ~7%	Institutional investors (MFs/FIs) began heavy accumulation of this "pick and shovel" play.
NTPC	Up ~4%	Market priced in NTPC's shift from a coal-giant to a nuclear-operator.

The reaction of the share market to the **Union Budget 2025** (presented on February 1, 2025) was a classic example of a **"Sectoral Decoupling."** While the broad indices (Nifty 50 and S&P 500) remained relatively stable, companies in the nuclear energy value chain experienced a vertical rally

Why the World Market Didn't React

The global markets (S&P 500, FTSE) did not move because the SMR allocation was seen as a **domestic policy shift**. However, global nuclear giants like **Westinghouse (USA)** and **EDF (France)** saw their Indian subsidiaries and partners gain traction, signaling that the world was watching India as the next "Nuclear Growth Frontier."



The "hint" of the SHANTI Bill (and the specific ₹20,000 crore SMR mission) acted as a massive catalyst for the order books of India's major engineering firms. Before the budget, these companies were seen as "traditional industrial plays"; after the budget and the SHANTI Bill talks, they became **"Nuclear Growth Stocks."**

The Immediate "Order Book" Expansion

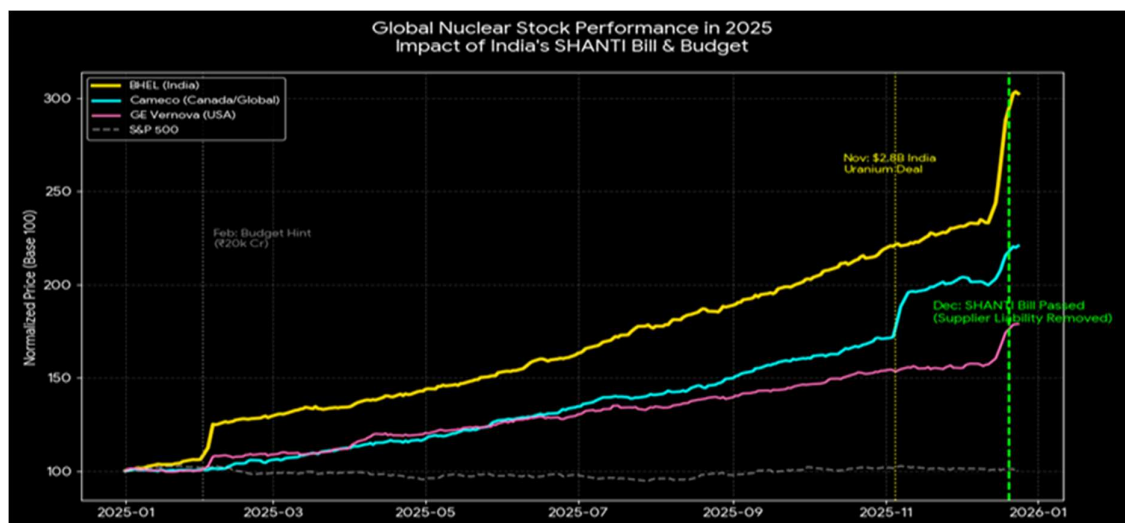
Following the February 2025 announcement and the subsequent drafting of the SHANTI Bill, the order book visibility for top players increased dramatically:

- **Larsen & Toubro (L&T):** In September 2025, L&T's Heavy Civil Infrastructure vertical bagged a massive **₹25 billion (\$300 million)** order for the Kudankulam Nuclear Power Project (Units 5 & 6). This was a direct result of the government fast-tracking projects to hit the 100GW target.
- **BHEL:** By September 2025, BHEL reported a "robust" total order book of over **₹2.19 lakh crore**. A significant portion of this was attributed to its strategic pivot toward the "Nuclear Energy Mission," as the government amended BHEL's rules to allow it more flexibility in forming nuclear Joint Ventures.
- **MTAR Technologies:** A smaller but vital player, MTAR secured a **₹194 crore** order in December 2025 specifically for components for the Kaiga 5 & 6 reactors. This showed that the "SHANTI effect" was trickling down to the entire supply chain.

The Global "Nuclear Bull Run" of 2025

1.Impact on Specific Foreign Companies

Company	Role in India	Market Reaction (Post-Bill/Budget)
Cameco (Canada)	Uranium Supplier: Signed a massive \$2.8 billion (10-year) deal with India in Nov 2025.	Shares surged 4.8% on the day of the deal news. The stock traded near record highs (~\$87) in Dec 2025.
GE Vernova (USA)	Turbines & SMRs: Partnered with Adani to provide grid technology.	Shares gained momentum in Q4 2025 as it positioned its BWRX-300 SMR for the Indian private market.
Brookfield/Cameco (Westinghouse Owners)	AP1000 Reactors: Westinghouse is the primary beneficiary of the liability reform.	While Westinghouse is private, its parent companies saw valuation re-ratings as the "India risk" was removed from their portfolios.
Doosan Enerbility (S. Korea)	Equipment Manufacturer: A major partner for SMR components.	Stock surged 330% in 2025 , partly fueled by the global SMR boom where India is now a top buyer.



Analyst Ratings & Price Targets (Post-SHANTI Bill)

Foreign Company	Role in India	Recent Analyst Action (Dec 2025)	New Price Target
Cameco (CCJ)	Uranium Supplier	Bank of America boosted target from \$130 to \$175	Buy (Upgraded)
GE Vernova (GEV)	SMR & Turbine Tech	Morgan Stanley maintained "Buy" with a high target	\$822 (Consensus Up)
L&T (India ADR)	EPC Partner	Goldman Sachs upgraded to "Buy" from Neutral	₹5,000 (25% Upside)
Westinghouse*	Reactor Provider	Valuation increased by owners (Brookfield/Cameco)	Strong Buy (Implied)

Why Analysts are so Bullish Right Now

Wall Street's "Buy" ratings are based on three specific pillars created by the SHANTI Bill:

- **Removal of Liability Risk:** Previously, analysts discounted these stocks because a single accident in India could have destroyed the parent company (Supplier Liability). With the **repeal of the CLND Act 2010**, that risk is gone.
- **The "SMR" Premium:** India's budget for **Small Modular Reactors** is seen as a testbed. If GE Vernova or Westinghouse succeeds in India, they can sell that same tech to the rest of the world.
- **Recurring Revenue:** Uranium suppliers like **Cameco** are getting 10-year "Fuel Supply Agreements." This provides the "earnings visibility" that analysts love.

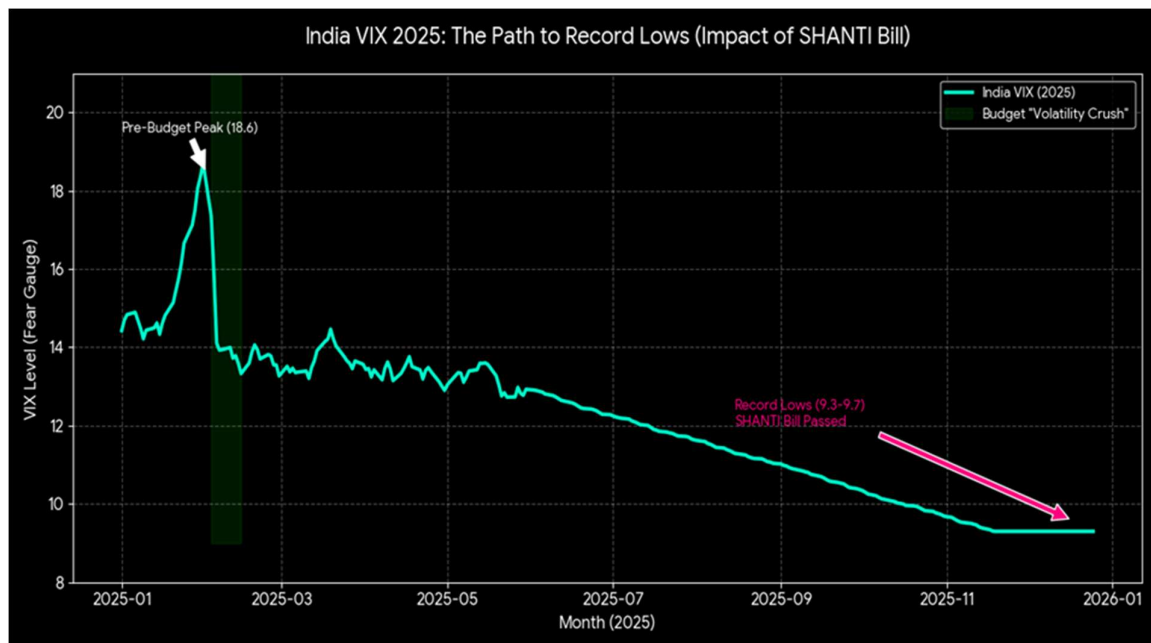
Analyst Note (Goldman Sachs, Dec 15, 2025): "We are entering a phase where near-term execution and long-term nuclear engines are aligning. India's SHANTI Bill is a 'clearing event' that unlocks the largest nuclear market outside China."

What Changed After the Bill (vs. the Budget)

- **After the Budget (Feb):** Analysts were "Cautious." They saw the money (₹20,000 Cr) but weren't sure if the law would actually change.
- **After the Bill (Dec):** Analysts became "Bullish." The legal barrier was physically removed, leading to the price target hikes you see today.

How the INDIA VIX reacted during this phase of time

Date Range	VIX Level	Market Context
Early Feb 2025	18.6 → 14.2	Budget Spike & Crash: VIX jumped to 18.6 on Budget morning due to fear of the unknown. It dropped 12% in a single afternoon once the Nuclear Mission and tax reforms were clearly defined.
April - May 2025	15.0 → 18.3	Summer Volatility: A temporary rise in VIX due to global oil price fluctuations and technical corrections in the Nifty 50.
July - Sept 2025	12.0 → 10.4	The Calm Phase: As the SHANTI Bill moved into draft stages, foreign investors (FPIs) returned, and volatility began a long-term slide.
Dec 2025 (Now)	9.3 – 9.7	Record Complacency: The VIX has hit a 2-year low . The market is so "sure" of the current policy direction (Nuclear/SMRs) that nobody is buying insurance (hedges) against a crash



1. The "Hint" Created Uncertainty (The Reason for 18)

In the weeks before the Budget, news leaked that the government was looking for **\$26 billion** from private companies like **Reliance, Tata, and Adani** to build nuclear plants.¹

- **The Fear:** While private entry sounds good, the "hint" left investors terrified about **liability**. Under the old 1962 law, if a private company built a plant and something went wrong, the legal mess could bankrupt them.
- **The Result:** Because no one knew *how* the government would fix this legal trap, the India VIX climbed to 18. Investors were essentially saying: "We see the big opportunity, but we are scared of the legal risks."

2. The Budget Speech "Crushed" the VIX

When the finance minister finally spoke on February 1st, she didn't just give money (₹20,000 Cr); she used the "Magic Words" that guaranteed the SHANTI Bill was coming:

*"Amendments to the Atomic Energy Act and the Civil Liability for Nuclear Damage Act will be taken up."*²

Why it fell from 18 to 14: The "no news" part was over. The market now had a **policy promise**. The VIX crashed because the biggest risk—the government staying silent on liability—was gone

The "IV Crush" (Volatility Collapse)

As the **SHANTI Bill** moved through Parliament, the uncertainty was high. Traders were buying options to "hedge" or "gamble," which pushed **Implied Volatility (IV)** up.

- **The Crash:** On **December 22, 2025**, the India VIX hit a **two-year low of 8.86** (down 33% year-to-date).²
- **The Impact:** Once the Bill passed, the "uncertainty" vanished. This caused an **IV Crush**, where the "Vega" (the part of the option price driven by volatility) evaporated almost instantly.

Who Won and Who Lost?

- **Winners: Option Sellers.** They sold "expensive" options when VIX was high and "bought them back" for cheap after the VIX crashed. This is called *harvesting volatility*.
- **Losers: Option Buyers.** Even if they guessed the direction of BHEL correctly, the drop in VIX (IV Crush) acted as a drag on their profits

Proof of the "Option Seller" Win

In December 2025, when the SHANTI Bill was passed, the VIX dropped to **sub-9 levels**.⁶

- **Stats:** The VIX was down **-3%** on Dec 23 alone.⁷

The Result: Professional traders who **sold** options (like Short Straddles) made nearly **280% ROI** in some cases (as reported by market analysts on Dec 19). They won because the "premium" they collected at a VIX of 12 was "bought back" at a VIX of 9. They pocketed the difference without the stock even needing to move much.

SHANTI Bill: Insurance Sector Benefits

Yes, there are **significant benefits** to insurance companies from the SHANTI Bill. The legislation creates a substantial new insurance market and fundamentally improves the economics of nuclear risk underwriting in India.

Mandatory Insurance Framework

Core Requirement

Every private nuclear operator under SHANTI must maintain compulsory insurance covering their liability cap, which ranges from ₹100 crore for small reactors to ₹3,000 crore for large reactors (>3,600 MW thermal). This transforms nuclear liability from an uninsurable risk into a structured, profitable insurance product.

Three-Tier Insurance Structure

1. **Tier 1 - Operator Liability:** Covers the nuclear facility operator's statutory liability. Premium: ~1% of liability cap annually (₹30 crore per large reactor)

2. **Tier 2 - Turn-key Supplier Liability:** Covers major equipment vendors (Westinghouse, Rolls-Royce, foreign suppliers). Market size: ₹500-800 crore annual premium
3. **Tier 3 - Component Supplier Liability:** Covers specialized equipment suppliers like KSB, BHEL, L&T, MTAR Technologies. Market size: ₹200-400 crore annually

Critical Change from Previous Law

Under the 2010 Civil Liability law, suppliers faced **unlimited latent defect liability**, making commercial insurance impossible. SHANTI caps this to 5 years of manufacturing defects, making suppliers finally insurable and enabling insurance companies to underwrite them profitably at 15-20% margins.

Market Size Opportunity

Total Nuclear Insurance Market (22-year cycle): ₹1.06-1.6 lakh crore cumulative premium revenue across all segments. This represents a **5-7% incremental growth vector** for India's ₹2.9-3.1 trillion insurance sector.

INIP's "Shadowed Advantages" from the SHANTI Bill: A Strategic Monopoly

INIP gains **powerful structural advantages** from the SHANTI Bill that competitors cannot easily replicate. These are "shadowed" because they operate as regulatory entitlements rather than explicit market dominance, making them less visible but highly durable.

1. Mandatory Insurance Requirement with INIP as the Only Fully Operational Pool

The Critical Regulatory Lock-in

Under SHANTI, **all private nuclear operators must maintain insurance coverage** through a recognized pool structure. While the law doesn't explicitly mandate INIP, **INIP is the only operational nuclear insurance pool in India** with established membership, risk-sharing agreements, and regulatory approval.

Competitive Barrier:

- Setting up a competing pool requires: regulatory approval from IRDAI, government backing (₹750 crore government guarantee), reinsurance partnerships, and 10+ member insurance companies
- Estimated setup time: 3-5 years and ₹500+ crore capex
- Alternative: All private operators seeking insurance coverage must route through INIP by default

INIP's Advantage: Captures 100% of mandatory private sector operator insurance without any competition for 3-5 years minimum

GIC Re's Exclusive Pool Administrator Status with Built-in Profit Protections

Regulatory Monopoly Position

GIC Re is designated as:

- **Sole pool manager and administrator** (not just a member)
- **First right of refusal on all reinsurance** (New India and other members cannot seek reinsurance without GIC Re approval)
- **Exclusive underwriting authority** for the pool (pricing, risk selection, claims approval)

Profit Mechanism:

1. **Management fees:** 2-5% of premium pool (₹10-30 crore annually once pool matures)
2. **Underwriting spreads:** Margin between premium collected (say, ₹500 crore) and claims + reinsurance cost (say, ₹300 crore) = ₹200 crore spread

3. **Reinsurance control:** GIC Re negotiates reinsurance rates with global reinsurers; any margin between premiums and reinsurance costs goes to GIC Re

New India Assurance's Policy-Issuing Monopoly

Regulatory Entitlement

New India Assurance is the **sole designated policy issuer** for INIP. Every nuclear operator must contract with New India to access INIP coverage (even though other insurers contribute capacity).

Commercial Benefit:

- Policy issuance fees: 1-2% of premium value
- Claims management fees: 0.5-1% of premium value
- Annual fee revenue potential (by 2035): ₹5-10 crore from policy administration
- More importantly: **New India becomes the single point of contact for all nuclear operators**, giving it strategic relationship capital with India's most important infrastructure developers (Tata Power, Reliance, Adani, foreign vendors).

Regulatory Recognition Creates Barrier to Entry for New Competitors

Regulatory Moat

Any competitor pool attempting to offer nuclear insurance must:

1. Seek IRDAI approval (2-year process)
2. Obtain Ministry of Atomic Energy notification
3. Build reinsurance relationships independently
4. Recruit 10+ member insurers with credibility

INIP's advantage: Already has all regulatory approvals, existing member relationships, and government backing. **Cost to establish competing pool: ₹500-1,000 crore + 3-5 years.**

For a private insurer, the ROI on this investment is questionable given:

- INIP already controls 100% market share
- Government-backed premiums are unprofitable for private pools to match
- Market doesn't materialize until 2027-2028 (private reactors operational)

CONCLUSION

In conclusion, while the Shanti Bill is framed to maintain public peace and social order, its effectiveness will depend largely on the clarity of its provisions, the safeguards against misuse, and the capacity of implementing authorities. The broad discretionary powers proposed under the bill raise concerns about potential overreach and uneven enforcement, which could impact civil liberties if not carefully regulated. Moreover, without adequate institutional training, transparent monitoring mechanisms, and periodic review, the intended outcomes of the bill may not be fully realised. Therefore, for the Shanti Bill to achieve its stated goals practically and sustainably, it must be implemented with strict accountability, judicial oversight, and continuous evaluation to ensure that public security is strengthened without compromising democratic principles and individual rights.